

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
23MX26 - Java Programming Laboratory
Hands-on Worksheet 1

1. Smart Electricity Bill Calculator

A city electricity department charges consumers based on their monthly usage:

Units Consumed Charge per Unit

First 100 ₹1.50

Next 100 ₹2.50

Above 200 ₹4.00

Additionally, add a **fixed meter charge of 75**.

Task:

Write a Java program that:

- Accepts consumer name and units consumed
- Calculates the total electricity bill
- Displays the final bill amount

Sample Output:

Consumer Name: Ravi

Units Consumed: 275

Total Bill Amount: 812.50

2. ATM Withdrawal Simulation

An ATM allows withdrawal under the following rules:

- Minimum balance to be maintained: 1000
- Withdrawal amount must be a multiple of 100
- Withdrawal amount should not exceed account balance

Task:

Write a Java program that:

- Accepts current balance and withdrawal amount
- Checks whether withdrawal is possible
- Displays updated balance or failure message

3. Online Shopping Discount System

An online store gives discounts based on purchase amount:

Purchase Amount Discount

< ₹1000 No discount

₹1000–₹4999 10%

₹5000–₹9999 20%

≥ ₹10000 30%

Task:

Write a Java program to:

- Accept purchase amount
- Calculate discount
- Display final payable amount

4. Railway Ticket Fare Calculator

Railway ticket fare depends on age:

Age Group Discount

Below 5 Free

5–12 50%

13–59 No discount

60+ 40%

Base ticket fare is 500.

Task:

Write a Java program that:

- Accepts passenger name and age
- Calculates ticket fare
- Displays ticket details

5. Student Result Processing System

A school evaluates students as follows:

- Pass mark: 40 in each subject
- Average $\geq 75 \rightarrow$ Distinction
- Average $\geq 60 \rightarrow$ First Class
- Average $\geq 50 \rightarrow$ Second Class
- Otherwise $\rightarrow$ Pass

Task:

Write a Java program that:

- Accepts marks in 3 subjects
- Checks pass/fail
- Displays result classification

6. Mobile Data Usage Alert System

A mobile network provider categorizes data usage:

Data Used (GB)	Message
≤ 1 GB	Normal Usage
>1 and ≤ 3 GB	Warning
>3 and ≤ 5 GB	Alert
>5 GB	Data Limit Exceeded

Task:

Write a Java program that:

- Accepts data usage in GB
- Displays appropriate alert message

7. Restaurant Billing System

A restaurant provides the following menu:

Item	Price
Burger	₹120
Pizza	₹250
Pasta	₹180

Rules:

- GST = 5%
- If bill amount > 1000 , apply 10% discount

Task:

Write a Java program that:

- Accepts quantity for each item
- Calculates total bill including GST and discount

8. Traffic Signal Simulator

Traffic signals behave as follows:

Signal Color	Action
Red	Stop
Yellow	Ready
Green	Go

Task:

Write a Java program that:

- Accepts signal color
- Displays appropriate action

9. Salary Slip Generator

A company pays employees as follows:

- HRA = 20% of basic salary
- DA = 10% of basic salary
- Tax = 5% of gross salary

Task:

Write a Java program that:

- Accepts employee name and basic salary
- Calculates gross salary and net salary
- Displays salary slip

10. Digital Attendance System

An employee is marked **Present** if:

- Login time is before or equal to 9:15 AM
Otherwise marked **Late**

Task:

Write a Java program that:

- Accepts login time (hour and minute)
- Displays attendance status